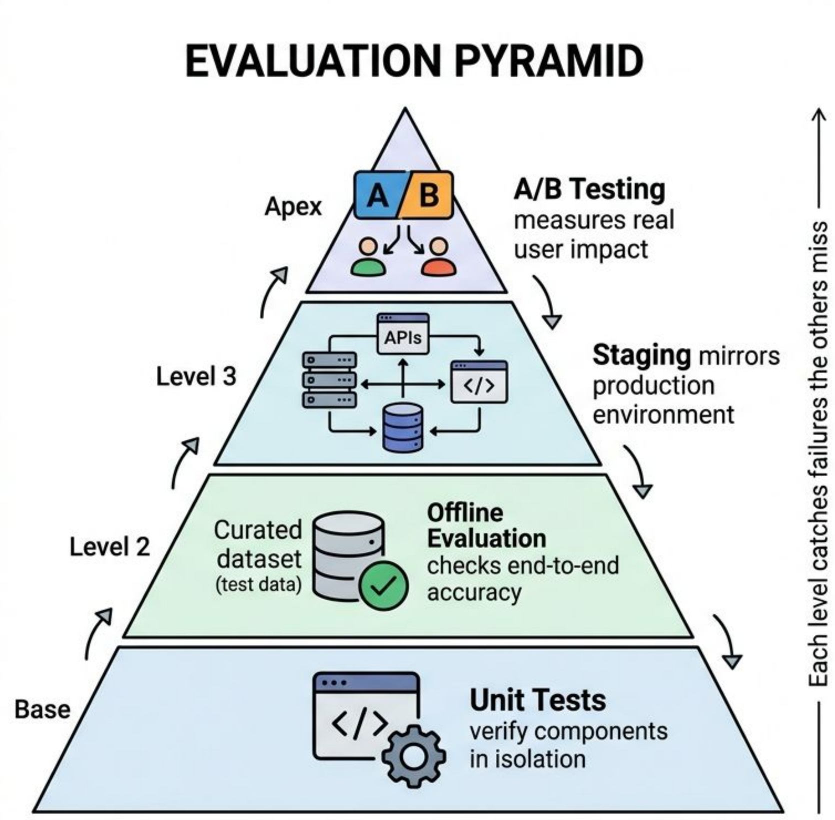
U.S. General Services Administration

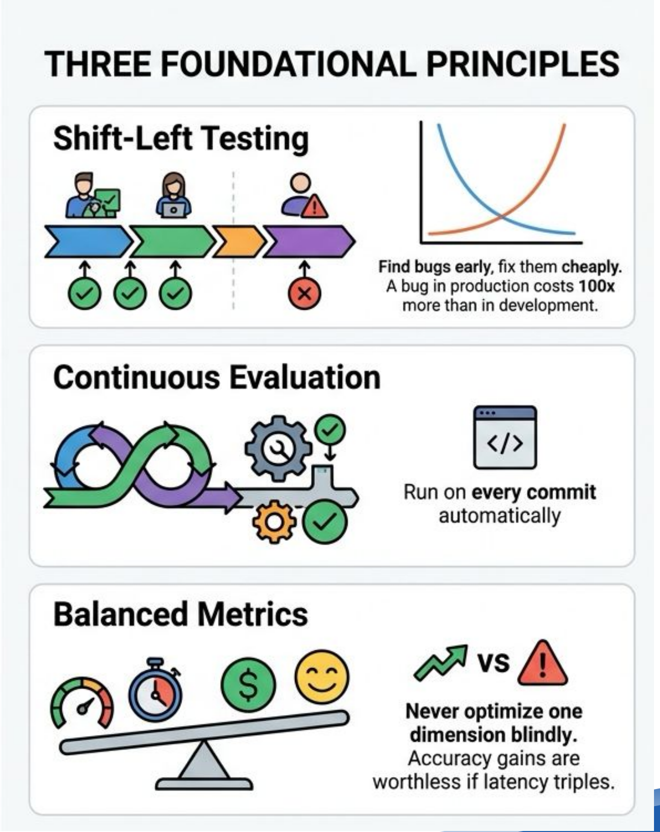
Chapter 3.1A: Implement Evaluation Pipelines and Task Benchmarks

US General Service Administration

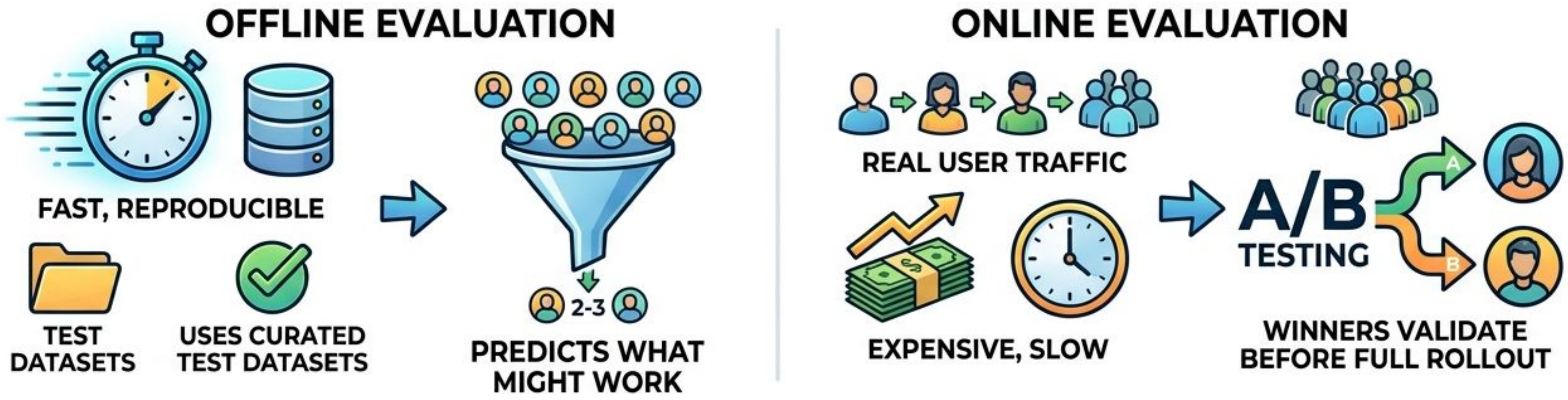
AI Community of Practice - March 2026

The Principles

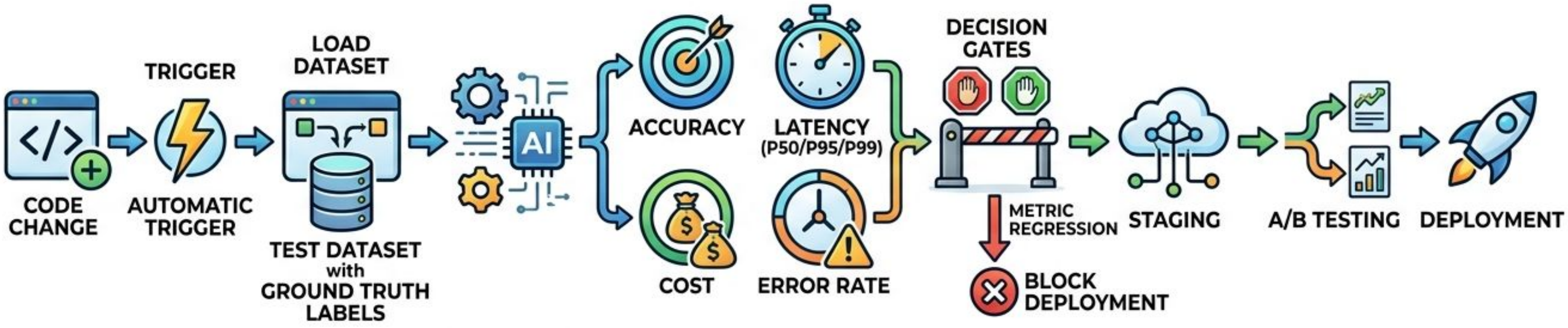




Offline vs. Online Evaluation

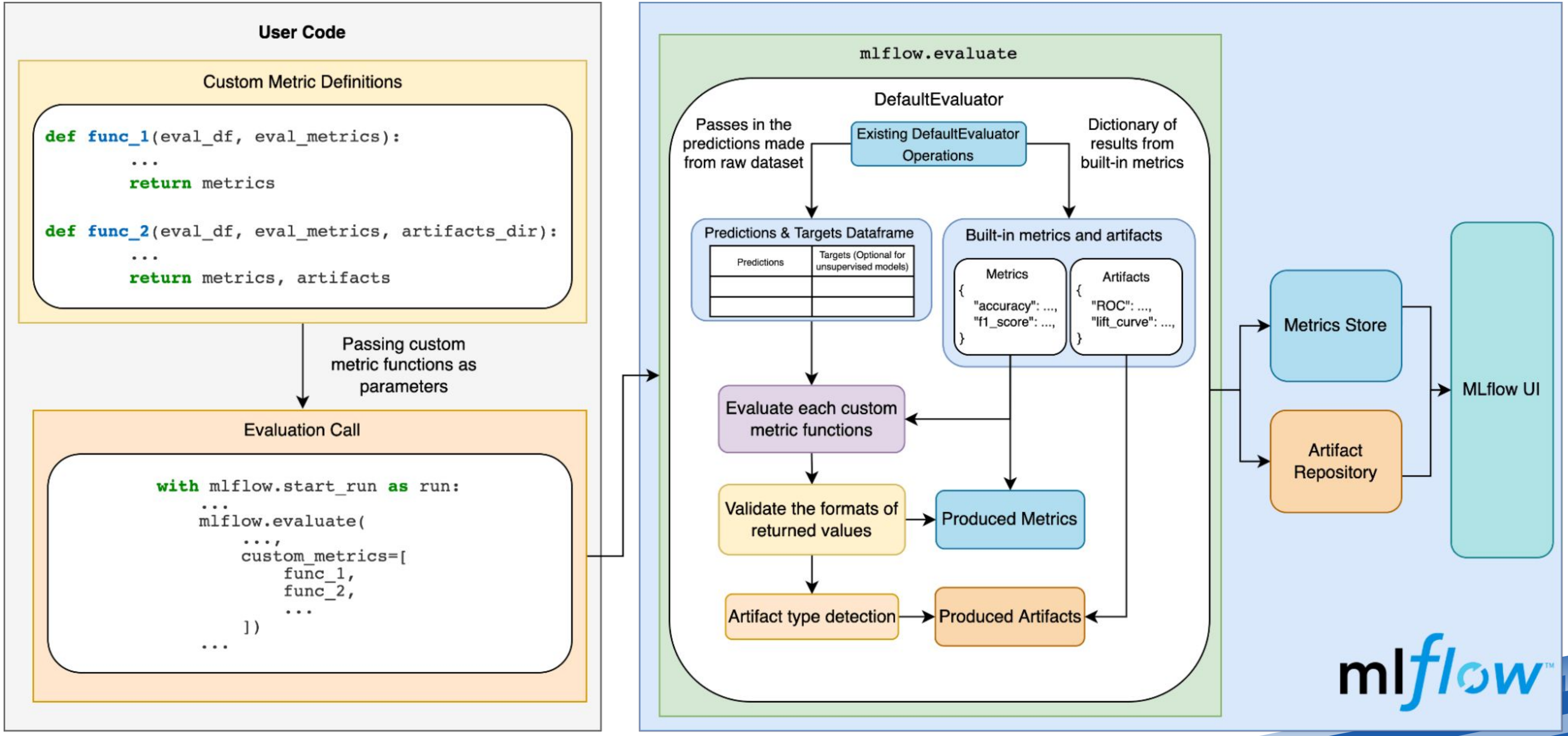


Evaluation Pipeline Architecture

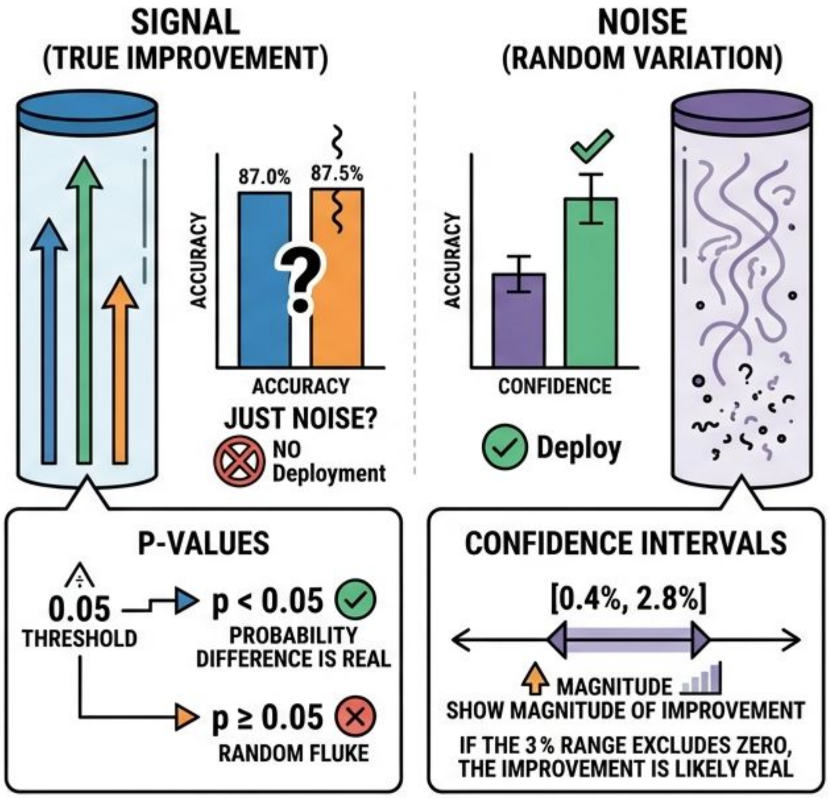


Read more: https://mlflow.org/blog/multiturn-evaluation/

Building an Offline Pipeline with MLflow

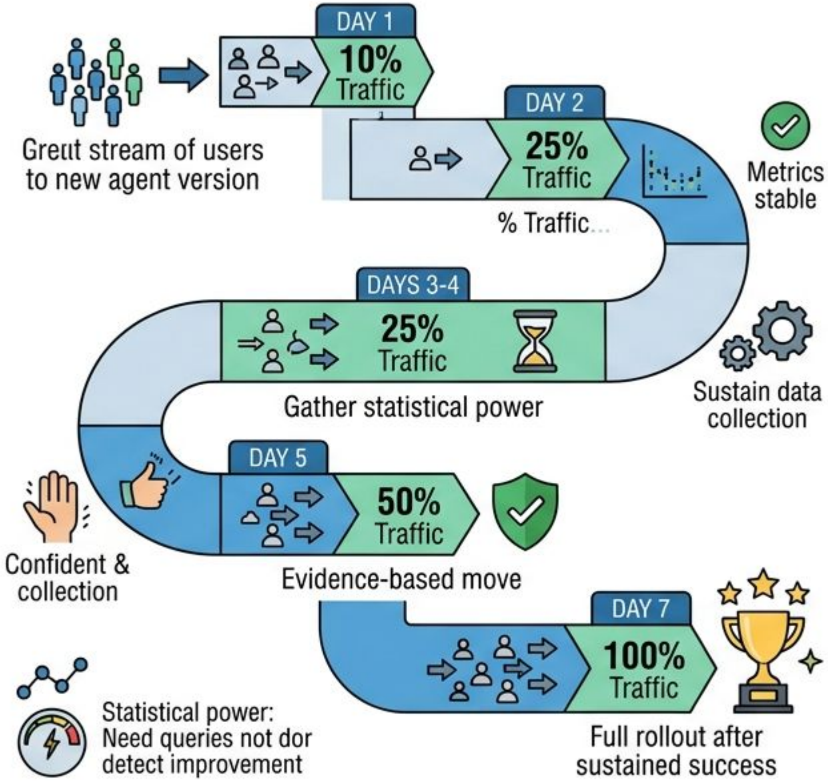


Statistical Significance



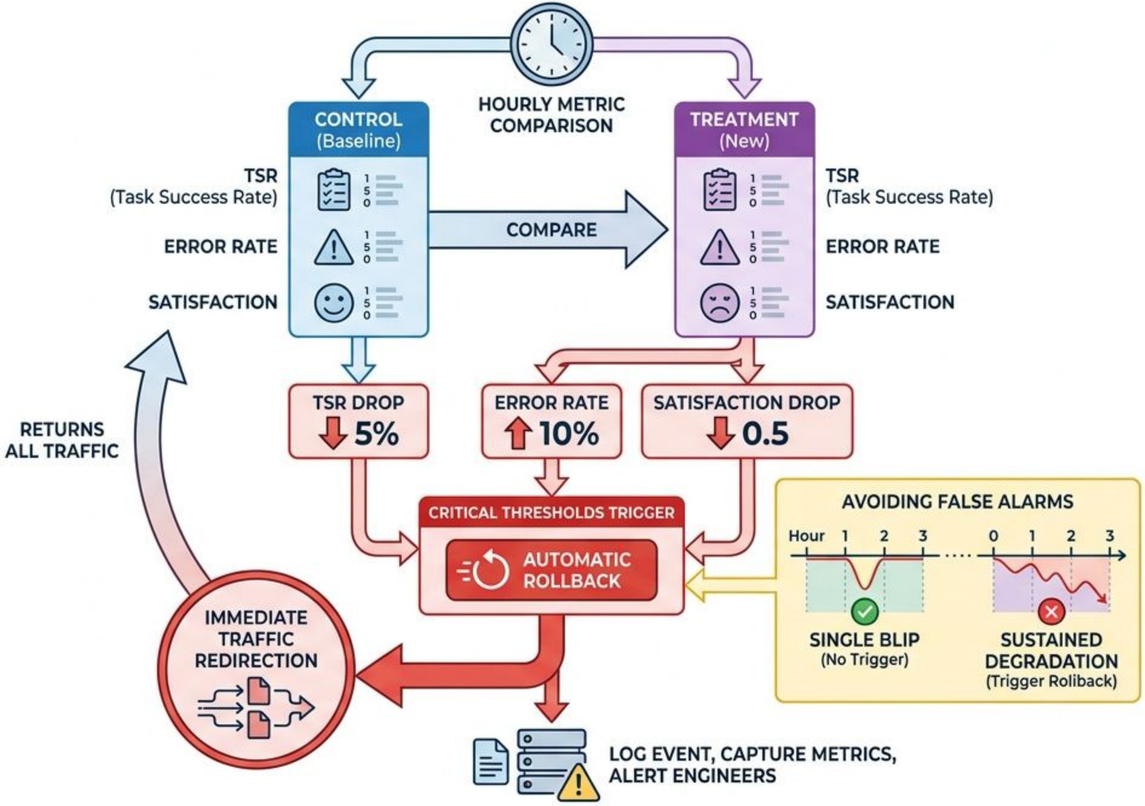
- Raw metric changes can be random variation
- 87.0% to 87.5% accuracy might just be noise
- P-values quantify probability difference is real
- $p < 0.05$ means less than 5% chance of random fluke
- Confidence intervals show magnitude of improvement

A/B Testing -- Progressive Rollout Strategy

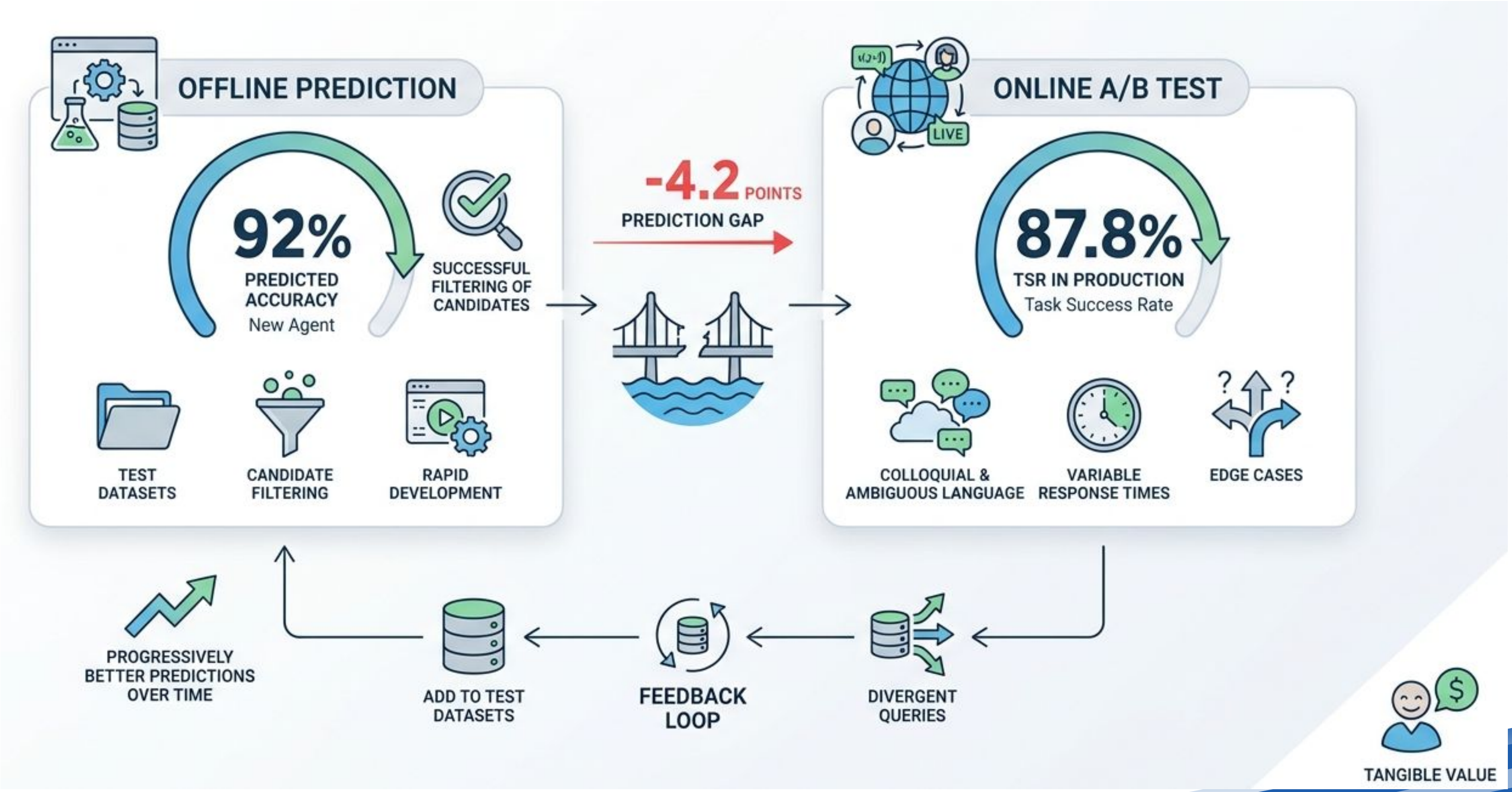


- Day 1: route 10% traffic to new agent version
- Day 2: scale to 25% if metrics hold steady
- Days 3-4: sustain and gather statistical power
- Day 5: confident move to 50% with evidence
- Day 7: full rollout to 100% after sustained success

Automatic Safeguards and Rollback



The Offline-to-Online Prediction Gap



Key Takeaways

- Task Success Rate is the primary agent reliability metric
- Evaluation pyramid: unit tests, offline, staging, A/B testing
- Shift-left testing reduces cost; continuous evaluation prevents regression
- Statistical testing distinguishes improvement from noise
- Always track multiple metrics to surface trade-offs